



## examGenome CheatSheet on Algebra

| Sl.no | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| 1     | <p><b>What is a map scale?</b></p> <p>Answer: A map scale shows the relationship between distances on the map and real distances on the ground.</p> <p>Explanation: For example, if the scale is 1 cm : 10 km, then 1 centimeter on the map equals 10 kilometers in real life.</p>                                                                                                                                                                       |
| 2     | <p><b>What are the two common ways to write a map scale?</b></p> <p>Answer: As a ratio (like 1:50,000) or as a statement (like 1 cm = 500 m).</p> <p>Explanation: The ratio form (1:50,000) means 1 unit on the map equals 50,000 of the same units in real life. The statement form uses real units like cm and km.</p>                                                                                                                                 |
| 3     | <p><b>What does a scale of 1:100,000 mean?</b></p> <p>Answer: It means 1 cm on the map equals 100,000 cm in real life (which is 1 km).</p> <p>Explanation: Since 100,000 cm = 1,000 meters = 1 kilometer, this scale is often used in road maps.</p>                                                                                                                                                                                                     |
| 4     | <p><b>Do map scales have units?</b></p> <p>Answer: The ratio form (like 1:25,000) has no units—it's the same unit on both sides. But statement scales use real units like cm and km.</p> <p>Explanation: In 1:25,000, it could be 1 cm : 25,000 cm or 1 inch : 25,000 inches—it's unitless. But '1 cm = 250 m' clearly uses centimeters and meters.</p>                                                                                                  |
| 5     | <p><b>Why are map scales important?</b></p> <p>Answer: They help you measure real distances using a ruler on a map.</p> <p>Explanation: Without a scale, you wouldn't know if two cities are 10 km or 100 km apart—just by looking at the map.</p>                                                                                                                                                                                                       |
| 6     | <p><b>What happens to the detail on a map when the scale number gets bigger (like 1:1,000,000 vs. 1:10,000)?</b></p> <p>Answer: A bigger scale number (like 1:1,000,000) means less detail (shows a larger area). A smaller scale number (like 1:10,000) shows more detail (smaller area).</p> <p>Explanation: Think: 'Large scale' = small area, more detail (like a neighborhood map). 'Small scale' = large area, less detail (like a world map).</p> |
| 7     | <p><b>How do you convert map distance to real distance using a scale?</b></p> <p>Answer: Multiply the map distance by the real distance represented by 1 unit on the map.</p> <p>Explanation: If 1 cm = 5 km and the map distance is 4 cm, then real distance = <math>4 \times 5 = 20</math> km.</p>                                                                                                                                                     |
| 8     | <p><b>What is a 'representative fraction' in map scales?</b></p> <p>Answer: It's the ratio form of a scale, like 1:50,000, where both numbers are in the same unit.</p> <p>Explanation: It's called a fraction because it's like <math>\frac{1}{50,000}</math>—showing how much the real world is reduced to fit on the map.</p>                                                                                                                         |

| <i><b>Sl.no</b></i> | <i><b>Content</b></i>                                                                                                                                                                                                                                                                                             |
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| 9                   | <p><b>Can you use inches or centimeters to measure on a map?</b></p> <p>Answer: Yes! But you must match the unit used in the scale statement.</p> <p>Explanation: If the scale says '1 inch = 4 miles,' use inches. If it says '1 cm = 2 km,' use centimeters.</p>                                                |
| 10                  | <p><b>What is the key rule to remember when using map scales?</b></p> <p>Answer: Always make sure your units match—and use the scale as a 'conversion rule' between map and real life.</p> <p>Explanation: Never skip the scale! Always check whether it uses cm, inches, km, or miles—and convert correctly.</p> |

